MID TERM EXAMINATION

Subject: Linked Lists, Stacks, Queues

Paper: Mid Term Examination - Set B | Duration: 3 Hours | Max Marks: 40 | Date: 11-03-2026

Total Questions: 11 | Answer all questions.

Section 1: MCQ (5 Questions - 10 Marks)

Q1. [2 marks] (Medium)

EXAM: Mid Term Examination | Course: Data Structures (CS101)

Q2. [2 marks] (Easy)

In a singly linked list, which part of a node typically stores the memory address of the subsequent node?

Options:
A) Data field B) Head pointer C) Next pointer D) Previous pointer

Q4. [2 marks] (Easy)

Which operation is used to add an element to the rear of a Queue?

Options:
A) Pop B) Dequeue C) Enqueue D) Peek

Q7. [2 marks] (Medium)

What is the expected behavior when an attempt is made to perform a pop operation on an empty stack?

Options:
A) The stack automatically duplicates the last item B) An underflow error typically occurs C) A random value is returned D) The stack resizes itself to accommodate a new element

Q11. [2 marks] (Medium)

If the primary requirement for a data structure is to efficiently retrieve the most recently added item, which of the following would be the most suitable choice?

Options:
A) Queue B) Array C) Singly Linked List D) Stack

Section 2: Short Answer (6 Questions - 30 Marks)

Q3. [5 marks] (Medium)

Explain the Last-In-First-Out (LIFO) principle as it applies to a Stack data structure. Provide a simple real-world analogy.

Q5. [5 marks] (Hard)

Describe the general iterative algorithm to reverse a singly linked list. You do not need to provide code, but explain the key pointers involved and how they are manipulated in each step.

Q6. [5 marks] (Easy)

Identify a common real-world scenario where a Queue data structure would be an appropriate choice for managing tasks. Briefly justify your answer.

Q8. [5 marks] (Medium)

Explain why searching for an element by its value (e.g., finding the 5th element) in a singly linked list is generally less efficient than in an array.

Q9. [5 marks] (Hard)

Describe how a circular queue addresses the issue of "wasted space" or "false full" conditions that can arise in a simple linear array-based queue implementation where elements are only added and removed from the ends.

Q10. [5 marks] (Hard)

Compare and contrast singly linked lists and doubly linked lists regarding the efficiency of deleting a specific node, given a pointer to that node.

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